

# Emanuele PIGANI



## PERSONAL DATA

PLACE AND DATE OF BIRTH: San Daniele del Friuli, Italy | 1<sup>st</sup> May 1995  
ADDRESS: Via Vigonovese 98/C, 35020 Padova, Italy  
STATUS: Married. Three children.  
PHONE: +39 334 1150305  
EMAIL: [emanuele.pigani@me.com](mailto:emanuele.pigani@me.com)

## RESEARCH

FROM APRIL 2025



Post-doc at Quantitative Life Sciences **International Center for Theoretical Physics**, Trieste

## EDUCATION

OCT 2021 - MAR 2025



Ph.D. at **Stazione Zoologica**, Napoli, and Laboratory of Interdisciplinary Physics, **University of Padova**, Padova

Supervisor: Dr. DANIELE IUDICONE

Co-supervisors: Prof. SAMIR SUWEIS, Prof. SANDRO AZAELE

Research topic: Modeling Macroecological Patterns in Marine Ecosystems: From Community structures to Gene Expressions

NOV 2020 - SEP 2021



Research Grant on STATISTICAL MECHANICS OF SPATIAL ECOSYSTEMS at the Laboratory of Interdisciplinary Physics, **University of Padova**, Padova

PI: Prof. AMOS MARITAN

JAN 2020 - OCT 2020



Research Grant on OPINION DYNAMICS at the ICT Center of **Fondazione Bruno Kessler**, Trento

PI: Dr. BRUNO LEPRI

OCT 2017 - OCT 2019



Master's Degree in PHYSICS, **University of Padova**, Padova

Thesis: *Analysis and modelling of information ecosystems*.

Advisor: Prof. SAMIR SUWEIS

Co-Advisors: Dr. CATERINA DE BACCO, Dr. CARLOS ALBERTO PLATA

FINAL GRADE: 110/110

GPA: 29/30

JUL 2019 - SEP 2019



Master Thesis Traineeship at the PHYSICS FOR INFERENCE AND OPTIMISATION GROUP, **Max Planck Institute for Intelligent Systems**, Tübingen

Supervisor: Dr. CATERINA DE BACCO

APR 2018 - SEPT 2018



Erasmus semester at the DEPARTMENT OF PHYSICS AND ASTRONOMY, **Ruprecht-Karls-Universität, University of Heidelberg**, Heidelberg

Theoretical Biophysics, Nonequilibrium Thermodynamics and Statistics, Computational Statistics and Data Analysis, Standard Model.

OCT 2014 - SEPT 2017



Bachelor's Degree in PHYSICS, **University of Padova**, Padova

Thesis: *Applications of Voter Model to Ecosystems*

Advisor: Prof. SAMIR SUWEIS

FINAL GRADE: 107/110

GPA: 28/30

## SELECTED PUBLICATIONS

|                                                 |                                                                                                                                                                                                                   |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Science Advances 10<br>(2024)                   | E. Pigani, B. Hay Mele, L. Campese, E. Ser-Giacomi, D. Iudicone, S. Suweis <i>Deviation from neutral species abundance distributions unveils geographical differences in the structure of diatom communities.</i> |
| PNAS 119 (2022)                                 | E. Pigani, D. Sgarbossa, S. Suweis, A. Maritan, S. Azaele <i>Delay effects on the stability of large ecosystems.</i>                                                                                              |
| Nature<br>communications 12<br>(2021)           | G. Cencetti, G. Santin, A. Longa, E. Pigani, A. Barrat, C. Cattuto, S. Lehmann, B. Lepri. <i>Digital proximity tracing on empirical contact networks for pandemic control.</i>                                    |
| Physical Review<br>Research 3, 013070<br>(2021) | C.A. Plata, E. Pigani, S. Azaele, V. Callejas, M.J. Palazzi, A. Solé-Ribalta, S. Meloni, J. Borge-Holthoefer, S. Suweis. <i>Neutral Theory for competing attention in social networks.</i>                        |

## CONFERENCES, TALKS AND WORKSHOPS

|                                                 |                                                                                                                                                          |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13 <sup>TH</sup> JUN 2024                       | SIFS Young seminars, Italian Society of Statistical Physics, Italy. <i>Invited talk</i>                                                                  |
| 23 <sup>RD</sup> - 28 <sup>TH</sup> JUN 2024    | Synergy Summer School: Complexity and emergence in marine ecosystems/seascape, Ischia, Italy.                                                            |
| 3 <sup>RD</sup> - 7 <sup>TH</sup> JUN 2024      | International Physics of Living Systems Annual Meeting, ICTP, Trieste, Italy. <i>Contributed talk</i>                                                    |
| 2 <sup>ND</sup> - 13 <sup>TH</sup> OCT 2023     | Advanced School on Quantitative Principles in Microbial Physiology: from Single Cells to Cell Communities, ICTP, Trieste, Italy. <i>Contributed talk</i> |
| 15 <sup>TH</sup> - 18 <sup>TH</sup> MAY 2023    | CEODOS and AtlantECO Workshop, Santiago, Chile. <i>Contributed talk</i>                                                                                  |
| 19 <sup>TH</sup> - 23 <sup>RD</sup> FEB 2023    | LiphLab Winter Workshop 2023, Gallio, Italy.                                                                                                             |
| 14 <sup>TH</sup> - 15 <sup>TH</sup> FEB 2023    | HFSP Kick-off meeting, Padova, Italy.                                                                                                                    |
| 1 <sup>ST</sup> - 3 <sup>RD</sup> OCT 2022      | AtlantECO meeting, Sesimbra, Portugal. <i>Contributed talk</i>                                                                                           |
| 7 <sup>TH</sup> - 10 <sup>TH</sup> JUN 2022     | Tara Oceans retreat, Niece, France. <i>Contributed talk</i>                                                                                              |
| 30 <sup>TH</sup> MAR - 2 <sup>ND</sup> APR 2022 | LiphLab Spring Workshop 2022, Asiago, Italy.                                                                                                             |
| 16 <sup>TH</sup> - 18 <sup>TH</sup> JUL 2020    | LiphLab Winter Workshop 2020, Folgaria, Italy.                                                                                                           |
| 1 <sup>ST</sup> - 3 <sup>RD</sup> JULY 2019     | First Italian Conference on Complex Systems, Bruno Kessler Foundation, Trento.                                                                           |
| 16 <sup>TH</sup> - 18 <sup>TH</sup> DEC 2018    | LiphLab Winter Workshop 2018, Folgaria, Italy.                                                                                                           |

## TEACHING EXPERIENCES

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|---------------------|------------------------------------------------------------------------------------------------------------------------------------|
| NOV 2021 - PRESENT  | Tenure Teacher of Physics and Math, MIM, Italy.                                                                                    |
| OCT 2023 - JUN 2024 | Teaching Assistant of Calculus and Physics, School of Agricultural Sciences and Veterinary Medicine, University of Padova, Padova. |
| OCT 2023 - JAN 2024 | Teaching Assistant of Models of Theoretical Physics, Department of Physics and Astronomy, University of Padova, Padova.            |
| OCT 2021 - DEC 2021 | Teaching Assistant of Calculus and Physics, School of Agricultural Sciences and Veterinary Medicine, University of Padova, Padova. |
| OCT 2020 - DEC 2020 | Teaching Assistant of Calculus and Physics, School of Agricultural Sciences and Veterinary Medicine, University of Padova, Padova. |
| NOV 2019 - DEC 2019 | High School Math teaching, ISIS "A. Malignani", Udine.                                                                             |

## IT SKILLS

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Advanced Knowledge: Python, Julia, Mathematica, C++, LINUX, ~~W~~TEX, GitHub  
Good Knowledge: ROOT, Fortran, LabView, Wordpress, R

## LANGUAGES

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ITALIAN: Native.  
ENGLISH: Fluent. C1 IELTS certificate (2021)  
GERMAN: Intermediate. C1 level for the Erasmus OLS language assessment (2018).

## AWARDS AND RELEVANT ACTIVITIES

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| SEP 2022            | <b>On board experience</b> on the Tara schooner during the oceanic cruise “Mission Microbiome”, from Senegal to Portugal.                                                                                |
| APR 2019            | <b>Student in charge</b> of the exhibition “The White Rose - Faces of a friendship”, sponsored by the University of Padova.                                                                              |
| MAY 2018 - OCT 2019 | <b>Student representative</b> , Physics Course, Physics and Astronomy Department, University of Padova, Padova.                                                                                          |
| NOV 2014 - APR 2016 | <b>Student representative</b> , “Commissione Paritetica”, School of Science, University of Padova, Padova.                                                                                               |
| 2011 - 2013         | <b>Student representative</b> , Consulta Provinciale degli Studenti, Udine.                                                                                                                              |
| 2012 - 2014         | <b>Captain</b> of his high school Math team.<br>1 <sup>st</sup> place at the Provincial Math Team Cup, Udine, 2014.<br>14 <sup>th</sup> place at the nationwide Italian Math Team Cup, Cesenatico, 2013. |
| FEB 2012            | <b>Honorable mention</b> at the nationwide literary competition “Colloqui Fiorentini”, in collaboration with MIUR, Firenze.                                                                              |

## NON-PROFIT EXPERIENCES

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| OCT 2021 - PRESENT  | <b>Tutoring</b> , Portofranco, Padova.<br>Volunteer as math and physics tutor for secondary school students.                 |
| OCT 2014 - OCT 2019 | <b>Customer service volunteer</b> , CUSL Nuova Vita, Padova.<br>Volunteer for a non-profit copy shop maintained by students. |
| OCT 2013 - JUN 2014 | <b>Tutoring</b> , Il Pellicano, Udine.<br>Volunteer as tutor for secondary school students.                                  |